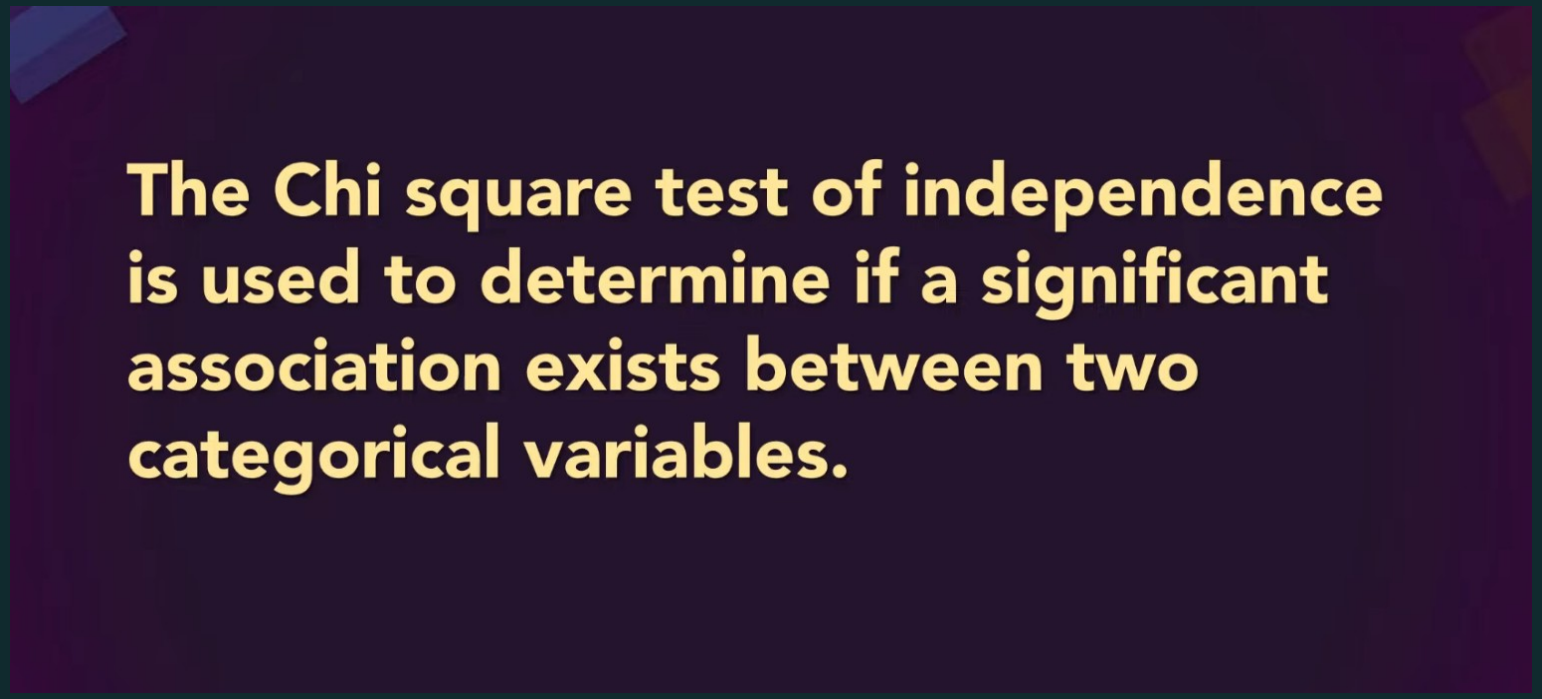
MBA BUSINESS STATISTICS

The Chi-Square Test of Independence



Anatomy of a contingency table

Age group	Uses AI: Yes	Uses AI: No	Total
Under 30	**312**	178	**490**
30–49	273	183	**456**
50 & above	224	270	**494**
Total	**809**	**631**	**1,440**

Also called a two-way table, cross-classification table or cross-tab

Cell

One observed count, O. Here: 312 under-30s use AI.

Marginal totals

Row and column totals, sitting in the margins.

Grand total, n

Everyone in the sample: 1,440.

3 × 2 table

r = 3 rows (age), c = 2 columns (AI usage). Rows are always named first.

The question, in plain English

H_0 · Null hypothesis

Age and AI usage are independent.

The share of AI users is the same in every age group. Any gaps in the sample are just chance.

H_1 · Alternative hypothesis

Age and AI usage are associated.

At least one age group uses AI at a different rate from the others.



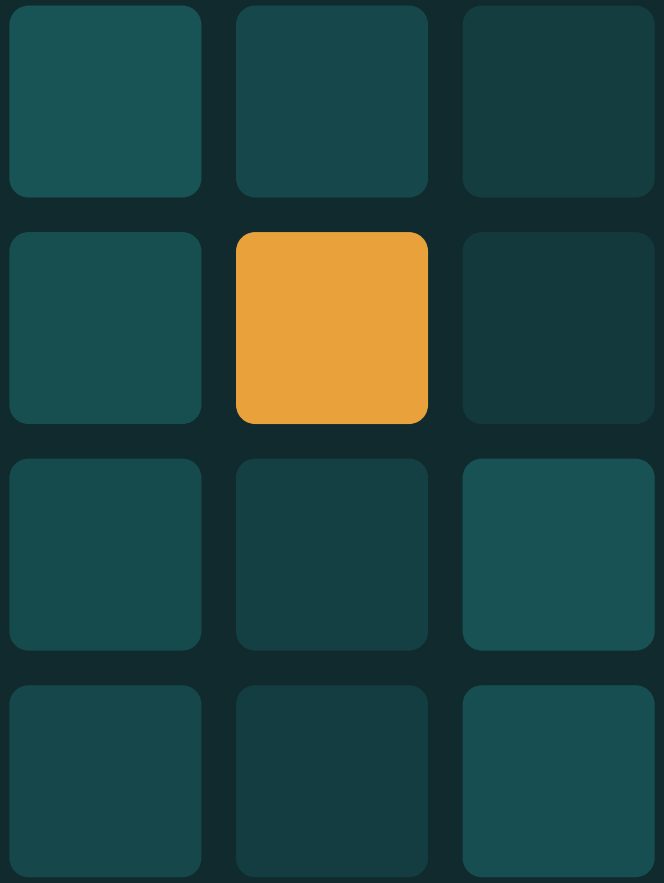
Why a manager cares: under H_0, one onboarding campaign fits everyone. Under H_1, the AI-tool budget should be split by age group. Same data, different spending decision.

The test compares what we observed with what we'd expect if H_0 were true.

02

The mechanics

The reference video's worked example, one step at a time: age group × AI usage, n = 1,440



Expected counts come from the margins

$$E = (\text{Row total} \times \text{Column total}) \div n$$

The count we'd expect in a cell if H_0 were true

Why this formula? Independence means probabilities multiply: P(row and column) = P(row) × P(column). Multiply by n and you get n × (R/n) × (C/n), which simplifies to R × C ÷ n.

Age group	Yes	No	Total
Under 30	**?**	178	**490**
30–49	273	183	456
50 & above	224	270	494
Total	**809**	**631**	**1,440**

Under 30 × Yes

490 × 809 ÷ 1,440

= 275.28

STEP 1 · EXPECTED COUNTS

Observed vs expected, side by side

Observed (O)

Age group	Yes	No	Total
Under 30	312	178	**490**
30–49	273	183	**456**
50 & above	224	270	**494**
Total	**809**	**631**	**1,440**

Expected if independent (E)

Age group	Yes	No	Total
Under 30	275.28	214.72	**490**
30–49	256.18	199.82	**456**
50 & above	277.53	216.47	**494**
Total	**809**	**631**	**1,440**

Totals match

Expected row and column totals equal the observed ones, give or take rounding. If they don't, recheck the arithmetic.

All E ≥ 5

The smallest expected count is 199.82, far above 5, so the test is valid here.

Add up the surprises: the χ^2 statistic

Cell	O	E	$(O - E)^2 \div E$
Under 30 · Yes	312	275.28	4.90
Under 30 · No	178	214.72	6.28
30–49 · Yes	273	256.18	1.10
30–49 · No	183	199.82	1.42
50 & above · Yes	224	277.53	**10.33**
50 & above · No	270	216.47	**13.24**
Total	**1,440**	**1,440**	**$\chi^2 = 37.26$**

$$\chi^2 = \Sigma\,(O - E)^2 \div E$$

Sum over every cell of the table

Squaring removes the sign, so over and under both count.

Dividing by E scales each gap to the size of its cell.

The 50 & above row supplies 23.6 of the 37.26 total.

Rounded parts add to 37.27, the unrounded total is 37.26. The video quotes both.

How many cells are actually free?

Age group	Yes	No	Total
Under 30	312	490 − 312	490
30–49	273	456 − 273	456
50 & above	809 − 585	494 − 224	494
Total	809	631	1,440

Free choice: pick any value

Forced by the totals

Once the margins are fixed, two cells decide the whole table.

$$df = (r - 1)(c - 1)$$
$$= (3 - 1)(2 - 1) = 2$$

Table size	df
2 × 2	1
3 × 3	4
4 × 3	6
5 × 4	12

STEP 5 · DECIDE

Two routes to the same decision

Critical value route

From a χ^2 table or Excel

Critical value (df = 2, α = .05)	**5.99**
Our statistic	**37.26**
Rule	**Reject H_0 if $\chi^2 > 5.99$**

37.26 > 5.99 → Reject H_0

p-value route

From software

$p = P(\chi^2 \geq 37.26)$	**≈ 0.000000008**
Significance level	**α = 0.05**
Rule	**Reject H_0 if $p < \alpha$**

p < 0.05 → Reject H_0

Conclusion in context: age and AI usage are associated. The pattern of AI use differs across age groups, and chance is a poor explanation for a gap this size.

11. A *Bloomberg Businessweek* subscriber study asked, "In the past 12 months, when traveling for business, what type of airline ticket did you purchase most often?" A second question asked if the type of airline ticket purchased most often was for domestic or international travel. Sample data obtained are shown in the following table.

	Type of Flight	
Type of Ticket	Domestic	International
First class	29	22
Business class	95	121
Economy class	518	135

a. Using a .05 level of significance, is the type of ticket purchased independent of the type of flight? What is your conclusion?
b. Discuss any dependence that exists between the type of ticket and type of flight.

12. A Deloitte employment survey asked a sample of human resource executives how their company planned to change its workforce over the next 12 months (*INC. Magazine,* February 2012). A categorical response variable showed three options: The company plans to hire and add to the number of employees, the company plans no change in the number of employees, or the company plans to lay off and reduce the number of employees. Another categorical variable indicated if the company was private or public. Sample data for 180 companies are summarized as follows.

	Company	
Employment Plan	**Private**	**Public**
Add Employees	37	32
No Change	19	34
Lay Off Employees	16	42

a. Conduct a test of independence to determine if the employment plan for the next 12 months is independent of the type of company. At a .05 level of significance, what is your conclusion?

b. Discuss any differences in the employment plans for private and public companies over the next 12 months.

14. A vehicle quality survey asked new owners a variety of questions about their recently purchased automobile (J.D. Power and Associates, March 2012). One question asked for the owner's rating of the vehicle using categorical responses of average, outstanding, and exceptional. Another question asked for the owner's education level with the categorical responses some high school, high school graduate, some college, and college graduate. Assume the sample data below are for 500 owners who had recently purchased an automobile.

	Education			
Quality Rating	**Some HS**	**HS Grad**	**Some College**	**College Grad**
Average	35	30	20	60
Outstanding	45	45	50	90
Exceptional	20	25	30	50

a. Use a .05 level of significance and a test of independence to determine if a new owner's vehicle quality rating is independent of the owner's education. What is the *p*-value and what is your conclusion?

b. Use the overall percentage of average, outstanding, and exceptional ratings to comment upon how new owners rate the quality of their recently purchased automobiles.

15. *The Wall Street Journal* Corporate Perceptions Study 2011 surveyed readers and asked how each rated the quality of management and the reputation of the company for over 250 worldwide corporations. Both the quality of management and the reputation of the company were rated on an excellent, good, and fair categorical scale. Assume the sample data for 200 respondents below applies to this study.

	Reputation of Company		
Quality of Management	**Excellent**	**Good**	**Fair**
Excellent	40	25	5
Good	35	35	10
Fair	25	10	15

a. Use a .05 level of significance and test for independence of the quality of management and the reputation of the company. What is the *p*-value and what is your conclusion?

b. If there is a dependence or association between the two ratings, discuss and use probabilities to justify your answer.

17. The National Sleep Foundation used a survey to determine whether hours of sleep per night are independent of age (*Newsweek,* January 19, 2004). A sample of individuals was asked to indicate the number of hours of sleep per night with categorical options: fewer than 6 hours, 6 to 6.9 hours, 7 to 7.9 hours, and 8 hours or more. Later in the survey, the individuals were asked to indicate their age with categorical options: age 39 or younger and age 40 or older. Sample data follow.

	Age Group	
Hours of Sleep	**39 or younger**	**40 or older**
Fewer than 6	38	36
6 to 6.9	60	57
7 to 7.9	77	75
8 or more	65	92

a. Conduct a test of independence to determine whether hours of sleep are independent of age. Using a .05 level of significance, what is the *p*-value and what is your conclusion?
b. What is your estimate of the percentages of individuals who sleep fewer than 6 hours, 6 to 6.9 hours, 7 to 7.9 hours, and 8 hours or more per night?

CASE 10 · CAMPUS PLACEMENT

Does specialisation decide placement?

The situation

A business school's MBA batch of 120 has finished final placements. The placement office recorded each student's specialisation and outcome.

The question A faculty member argues that specialisation decides placement. Test it, but check the assumptions first.

Your task: state H_0 and H_1, check assumptions, compute E and χ^2, decide at $\alpha = 0.05$, and name the cells driving the result.

Observed counts

Specialisation	Placed	Not placed	Total
Marketing	30	5	**35**
Finance	35	5	**40**
HR	16	4	**20**
Business Analytics	23	2	**25**
Total	**104**	**16**	**120**

Illustrative data · $\alpha = 0.05$

Think about it
Compute the expected 'Not placed' counts before anything else. What do you notice?

CHOOSING THE TEST

Which chi-square test fits your question?

Goodness of fit

One variable against a claimed distribution

Example
Are complaints spread evenly across the week?

$$df = k - 1$$

Independence

One sample, classified on two variables

Example
Is age linked to AI usage?

$$df = (r - 1)(c - 1)$$

Homogeneity

Separate samples from each group, one variable

Example
Do three cities share the same payment mix?

$$df = (r - 1)(c - 1)$$

Independence and homogeneity use identical arithmetic; they differ in how the sample was drawn. Paired yes/no data on the same people → McNemar's test.

SOFTWARE

Running it in Excel, Python, R and SPSS

Excel

```
=CHISQ.TEST(observed, expected)  → p
=CHISQ.INV.RT(0.05, df)  → cutoff
=CHISQ.DIST.RT(chi2, df)  → p
```

*Build the expected table first with =R*C/n.*

Python (SciPy)

```
from scipy.stats import chi2_contingency
chi2, p, dof, exp = chi2_contingency(
    table, correction=False)
```

correction=False matches hand calculations.

R

```
res <- chisq.test(tbl, correct = FALSE)
res$expected   # E table
res$stdres     # adjusted residuals
```

Yates is on by default for 2 × 2 tables.

SPSS

```
Analyze → Descriptive Statistics →
Crosstabs → Statistics: Chi-square, Phi
and Cramér's V → Cells: Adjusted std.
```

One dialog gives the whole expert package.